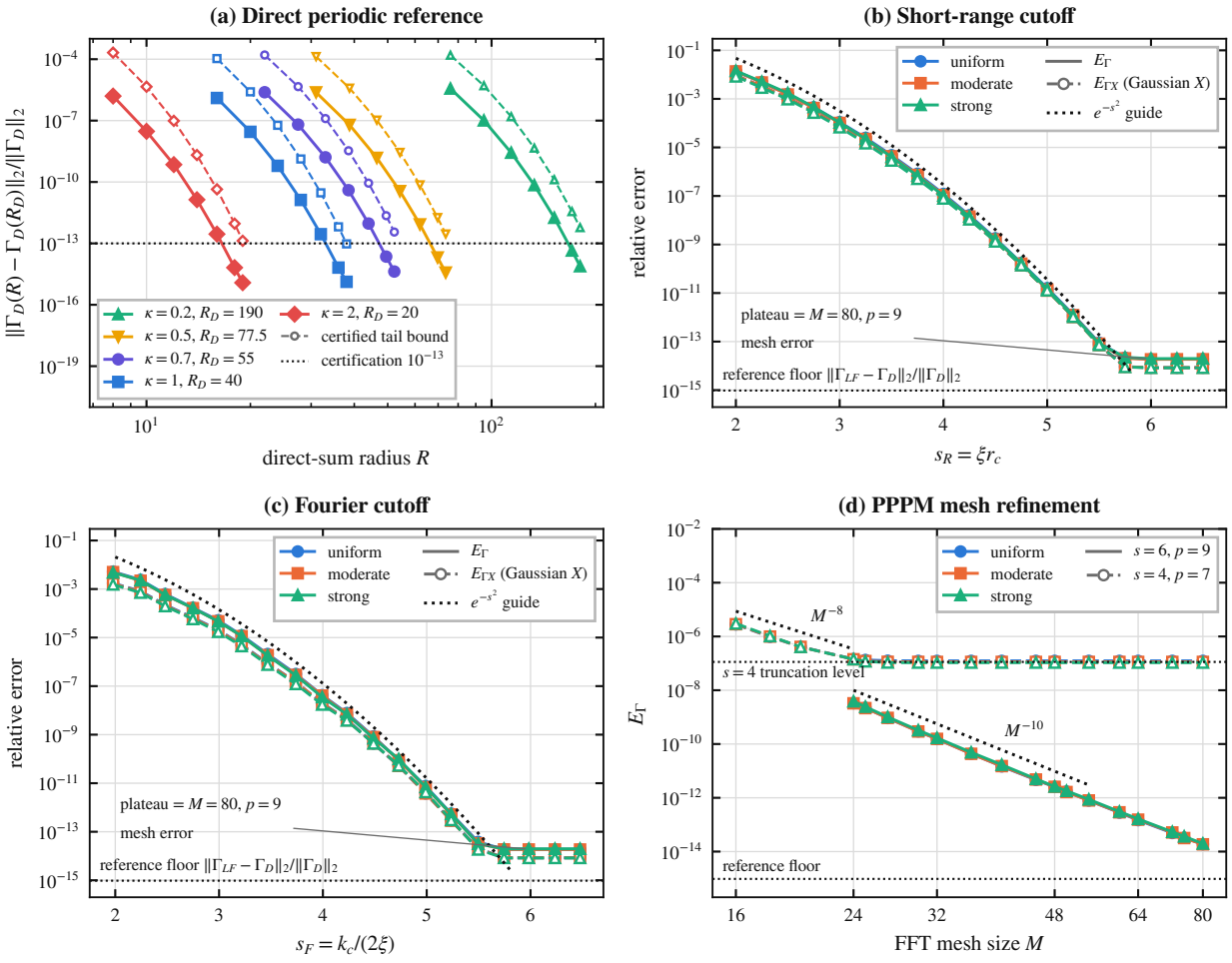
(a) Direct periodic reference
$\|\Gamma_D(R) - \Gamma_D(R_D)\|_2/\|\Gamma_D\|_2$
direct-sum radius R
$\kappa = 0.2$, $R_D = 190$
$\kappa = 0.5$, $R_D = 77.5$
$\kappa = 0.7$, $R_D = 55$
$\kappa = 1$, $R_D = 40$
$\kappa = 2$, $R_D = 20$
certified tail bound
certification 10^{-13}
(b) Short-range cutoff
relative error
$s_R = \xi r_c$
uniform
moderate
strong
E_Γ
$E_{\Gamma X}$ (Gaussian X)
e^{-s^2} guide
plateau $= M = 80$, $p = 9$
mesh error
reference floor $\|\Gamma_{LF} - \Gamma_D\|_2/\|\Gamma_D\|_2$
(c) Fourier cutoff
relative error
$s_F = k_c/(2\xi)$
uniform
moderate
strong
E_Γ
$E_{\Gamma X}$ (Gaussian X)
e^{-s^2} guide
plateau $= M = 80$, $p = 9$
mesh error
reference floor $\|\Gamma_{LF} - \Gamma_D\|_2/\|\Gamma_D\|_2$
(d) PPPM mesh refinement
E_Γ
FFT mesh size M
uniform
moderate
strong
$s = 6$, $p = 9$
$s = 4$, $p = 7$
M^{-8}
M^{-10}
$s = 4$ truncation level
reference floor